(i) Both dice first show a six on throw r:

$$q^{2r-2}p^2$$

Exactly one die first shows a six on throw s, the other first shows a six on throw r:

$$q^{2s-2} \cdot 2pq \cdot q^{r-s-1} \cdot p = 2p^2q^{r+s-2}$$

$$\begin{aligned} P(R=r) &= p^2q^{2r-2} + \sum_{s=1}^{r-1} 2p^2q^{r+s-2} \\ &= p^2q^{2r-2} + 2p^2q^{r-1} \cdot \frac{1-q^{r-1}}{1-q} \\ &= pq^{r-1}(2-q^{r-1}-q^r) \end{aligned}$$

$$\begin{aligned} E(R) &= \sum_{r=1}^{\infty} r \cdot pq^{r-1}(2-q^{r-1}-q^r) \\ &= p\left(2\sum_{r=1}^{\infty} rq^{r-1} - \sum_{r=1}^{\infty} rq^{2r-2} - \sum_{r=1}^{\infty} rq^{2r-1}\right) \\ &= p\left(\frac{2}{(1-q)^2} - \frac{1}{(1-q^2)^2} - \frac{q}{(1-q^2)^2}\right) \\ &= \frac{3-2p}{p(2-p)} \end{aligned}$$

(ii)

$$m = \frac{3-2p}{p(2-p)}$$

$$\implies mp^2 - (2m+2)p + 3 = 0$$

$$p = \frac{m+1 \pm \sqrt{m^2-m+1}}{m}$$

$$\because 0 < p \leq 1$$

$$\therefore p = \frac{m+1-\sqrt{m^2-m+1}}{m}$$

This solution was submitted by **Jon**

Published and refined by **Jon**

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